



Brier: Confidence-Calibrated Slashing for On-Chain AI Decision Accountability

A research proposal and v0 implementation report

Version v0 (prototype)

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Repository github.com/Sagexd08/Brier

This is a mechanism and a measured prototype, not a deployable protocol.

Only the calibration head is zero-knowledge proved — its input logit is unverified,

dispute resolution rests on an admin-appointed committee, and the figures reported here come from a local chain. Section 8 states each limitation in full.

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Abstract

Accountability mechanisms for automated decisions penalise error at a confidence-independent rate, so an operator is never punished for asserting 99% confidence in a decision it believes at 60%. We propose slashing staked collateral by the **Brier score** over (reported confidence, adjudicated outcome). Because that score is strictly proper, expected loss is minimised exactly when the operator reports its true subjective probability, making honest confidence reporting loss-minimising rather than good faith. We report a measured prototype: across 10 pinned seeds, temperature scaling cuts Expected Calibration Error from 0.1853 ± 0.0248 to 0.0870 ± 0.0218 on held-out UCI German Credit data, in every seed; an EZKL/halo2 circuit proves the calibration head in 2.09 s, verified on-chain for 684,696 gas. This is a mechanism and a prototype, **not a deployable protocol**: only the calibration head is proved, its input logit unverified, dispute resolution one admin key, no unbonding period.

1. Motivation

1.1 The regulatory setting

Automated credit decisions are moving into a compliance regime that demands records rather than assurances.

Under the EU AI Act, credit scoring is a high-risk application, and Article 26 places operational duties on **deployers** of such systems. Article 26(5) requires deployers to "monitor the operation of the high-risk AI system on the basis of the instructions for use" and to notify providers of risks or serious incidents without undue delay. Article 26(6) requires deployers to "keep the logs automatically generated by that high-risk AI system ... for a period appropriate to the intended purpose ... of at least six months." Article 86 establishes a right to explanation of individual decision-making. Article 26 becomes applicable on 2 December 2027 for most high-risk systems, with an extended deadline of 2 August 2028 for systems in employment, education, and access to essential services [1].

These provisions mandate *logging and explanation*. They do not establish that the logs are truthful, nor do they price the difference between a wrong decision made tentatively and a wrong decision made with asserted certainty. A deployer that logs a confidently wrong decision is compliant with Article 26(6).

A note on the Indian setting, correcting a common assumption. The RBI Digital Lending Directions, 2025 are frequently cited as imposing algorithmic transparency duties. They do not. The RBI Working Group recommended that regulated entities "document the rationale for algorithmic features and conduct regular algorithmic audits," but these recommendations were incorporated into neither the 2022 Guidelines nor the 2025 Directions, which remain silent on algorithmic fairness and transparency [2]. We therefore cite India as an example of a **regulatory gap** rather than a regulatory driver. We were unable to locate a provision in the 2025 Directions imposing model-explainability obligations, and we do not assert one exists.

The honest framing is that regulation currently creates demand for *auditable records of automated decisions* and does not yet create demand for *calibrated confidence*. This proposal argues the latter is the more useful object, not that it is presently required.

1.2 The gap in existing mechanisms

Decentralised protocols already slash staked collateral for misbehaviour, and already run decentralised claim adjudication. Neither weights the penalty by the reporter's own stated confidence.

- **Oracle slashing.** Chainlink slashes node operators for failing on-chain service-level obligations at a **fixed 700 LINK per valid alerting event** on the ETH/USD data feed, with the alerter rewarded 7,000 LINK [3]. The penalty is a constant: it is identical whether the operator was marginally or egregiously wrong, and oracles do not report confidence at all.
- **Decentralised insurance.** Nexus Mutual resolves claims by stake-weighted member voting, with claim assessors staking NXM and rewarded for voting with the outcome [4]. The mechanism adjudicates *whether a claim is valid*; it does not score how confident anyone was.
- **On-chain proper scoring rules.** These do exist. Foresight Arena (arXiv:2605.00420) scores AI forecasting agents on Polymarket questions using the Brier score, with commitments and scores recorded on Polygon PoS via smart contracts [5].

The precise delta is therefore narrower than "nothing like this exists," and we state it as such: ****proper scoring rules have been applied on-chain to *reward* forecasters in prediction-market settings; we are not aware of prior work applying one to *slash staked collateral* of a model operator under dispute, with the scored quantity produced by a zero-knowledge-proved calibration step.**** Foresight Arena is reward-based, involves no staked collateral at risk, no slashing, and no zero-knowledge proofs [5]. We claim novelty in the combination, not in any component.

2. Related work

2.1 zkML and verifiable inference

EZKL compiles computational graphs supplied in ONNX format into arithmetic circuits and produces zk-SNARK proofs of correct model execution, using a modified halo2 with Plonkish arithmetisation; anyone holding the verifying key can check the proof [6]. This prototype uses EZKL 23.0.5 directly. The broader design question EZKL raises — that proving cost scales with the proved subgraph — is what motivates proving a small calibration head rather than a full classifier, and is examined empirically in §6.3.

2.2 Decentralised oracle staking and slashing

Chainlink's staking design uses on-chain service-level agreements, community alerting, and fixed-size slashing penalties [3]. Nexus Mutual's claim assessment uses stake-weighted voting with a quorum, escalation to a full member vote, and reward for majority-aligned voting [4]. Kleros provides general decentralised arbitration: jurors are drawn from PNK stakers, self-select into sub-courts, vote on evidence, and are rewarded for voting with the majority under a Schelling-point incentive, with appeals heard by successively larger juries [7]. Kleros is the most plausible substitute for this prototype's trusted administrator (§8).

2.3 Calibration in machine learning

Guo et al. show that modern neural networks are poorly calibrated, and that temperature scaling — a single-parameter extension of Platt scaling that divides logits by a learned $T > 0$ fitted on held-out data — is surprisingly effective at restoring calibration [8]. Temperature scaling is the primary calibration method here, chosen both for its empirical strength and because a single-parameter map is trivially cheap to arithmetise.

2.4 Proper scoring rules

Brier introduced the mean-squared-error score for probability forecasts of binary outcomes [9]. Gneiting and Raftery develop the general theory: a scoring rule is *proper* if the forecaster maximises expected score by issuing its true predictive distribution F rather than any $G \neq F$, and *strictly proper* if that maximum is unique [10]. The Brier score is strictly proper, which is the sole property this mechanism depends on.

3. Core contribution

Claim (testable). Defining the slash on a disputed decision as $slash = S \cdot (c - o)^2$, where S is staked collateral, $c \in [0,1]$ the operator's reported confidence that its decision was correct, and $o \in \{0,1\}$ the adjudicated outcome, yields a mechanism in which reporting the operator's true subjective probability minimises expected loss — so honest confidence reporting is the loss-minimising strategy rather than an assumption — and this is implementable today, with the confidence-producing step proved in zero knowledge at 2.09 s on a commodity CPU and verified on-chain for 684,696 gas.

Scope conditions, stated as part of the claim rather than as caveats. The properness argument holds *conditional* on the adjudicated outcome being truthful, and on dispute selection being independent of the reported confidence. Neither condition is guaranteed by this prototype. Specifically:

1. **Input-logit provenance is unproven.** The circuit takes the base model's logit as an unverified public input. An operator that fabricates the logit obtains a proof that verifies. The proof binds the calibration step, not the pipeline (Figure C, tier 1; `ThreatModel.t.sol::test_tier1_marginIsUnverifiedOperatorSuppliedInput`).
2. **Off-chain misreporting before proving is not prevented.** Nothing forces the operator to feed the circuit the logit its deployed model actually produced.
3. **Selective disputes break the properness argument.** If only rejections are ever disputed, the operator faces an asymmetric loss surface, and the dominant strategy under that surface is not analysed here.

The mechanism is proper; the *system* is proper only to the extent that these conditions hold, and in vo they do not.

4. System design

4.1 Figure A — Component architecture

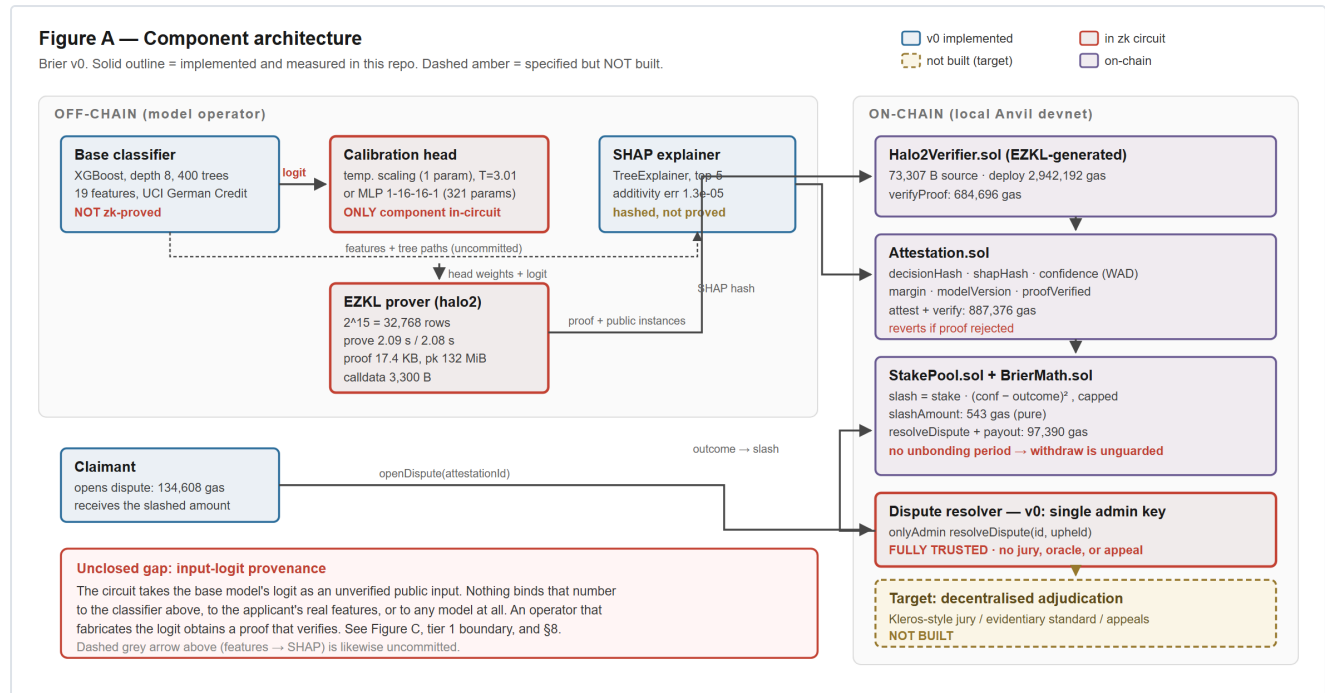


Figure A — Component architecture

Solid outlines mark components implemented and measured in the repository; dashed amber marks specified-but-unbuilt. Red marks the single component inside the zk circuit. The base classifier and the SHAP explainer sit outside the circuit by design; the SHAP vector is hash-committed as evidence only.

4.2 Figure B — Protocol sequence

Rendered below for PDF export; the source is inline Mermaid (and `figures/figure-b-sequence.mmd`) so it also renders natively on GitHub.

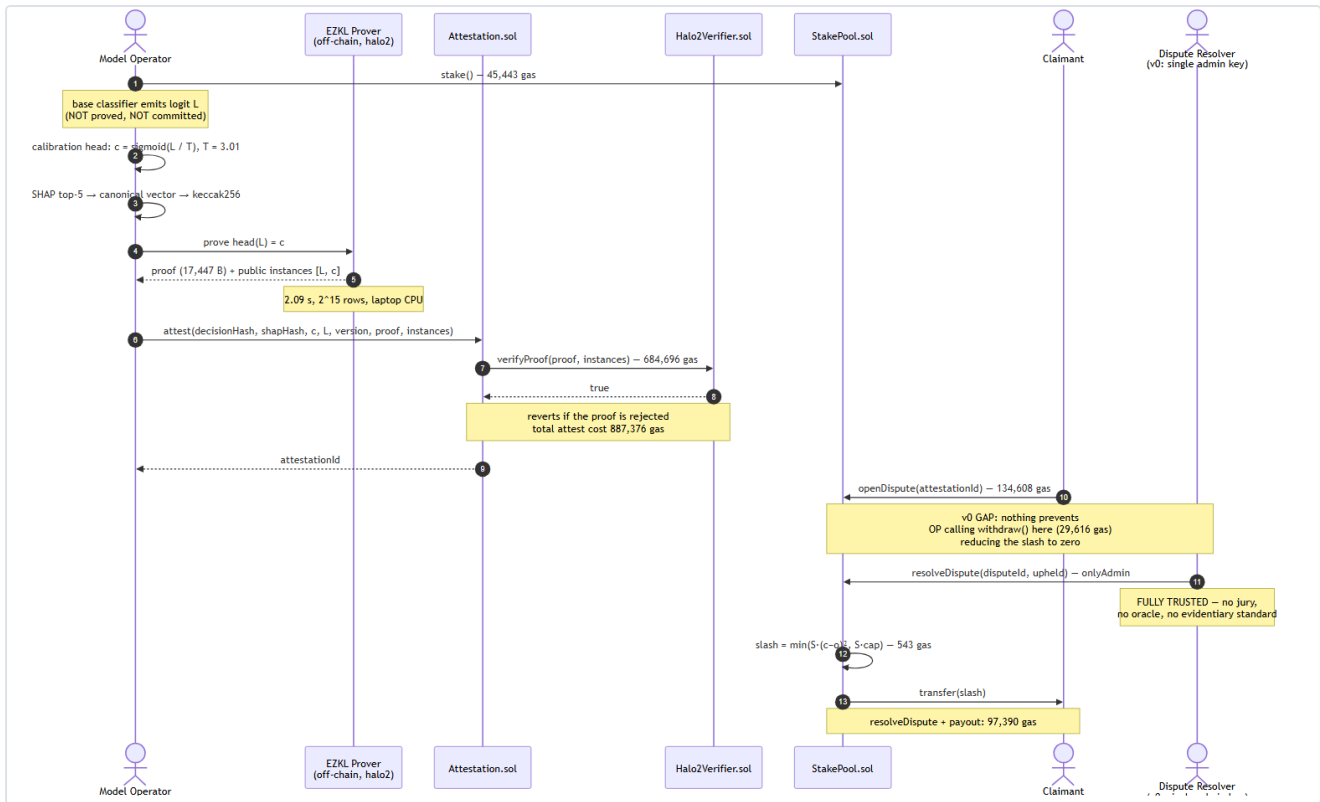


Figure B — Protocol sequence

```
sequenceDiagram
    autonumber
    actor OP as Model Operator
    participant PR as EZKL Prover<br/>(off-chain, halo2)
    participant AT as Attestation.sol
    participant VF as Halo2Verifier.sol
    participant SP as StakePool.sol
    actor CL as Claimant
    actor AD as Dispute Resolver<br/>(v0: single admin key)

    OP->>SP: stake() - 45,443 gas
    Note over OP: base classifier emits logit L<br/>(NOT proved, NOT committed)

    OP->>OP: calibration head:  $c = \text{sigmoid}(L / T)$ ,  $T = 3.01$ 
    OP->>OP: SHAP top-5  $\rightarrow$  canonical vector  $\rightarrow$  keccak256
    OP->>PR: prove head(L) = c
    PR-->>OP: proof (17,447 B) + public instances [L, c]
    Note over PR: 2.09 s,  $2^{15}$  rows, laptop CPU

    OP->>AT: attest(decisionHash, shapHash, c, L, version, proof, instances)
    AT->>VF: verifyProof(proof, instances) - 684,696 gas
    VF-->>AT: true
    Note over AT,VF: reverts if the proof is rejected<br/>total attest cost 887,376 gas
    AT-->>OP: attestationId

    CL->>SP: openDispute(attestationId) - 134,608 gas
    Note over SP,AD: v0 GAP: nothing prevents<br/>OP calling withdraw() here (29,616 gas)<br/>reducing the slash to zero

    AD->>SP: resolveDispute(disputeId, upheld) - onlyAdmin
    Note over AD: FULLY TRUSTED - no jury,<br/>no oracle, no evidentiary standard
    SP->>SP: slash =  $\min(S \cdot (c-o)^2, S \cdot \text{cap})$  - 543 gas
    SP->>CL: transfer(slash)
    Note over SP,CL: resolveDispute + payout: 97,390 gas
```

4.3 Figure C — Trust boundaries and threat model

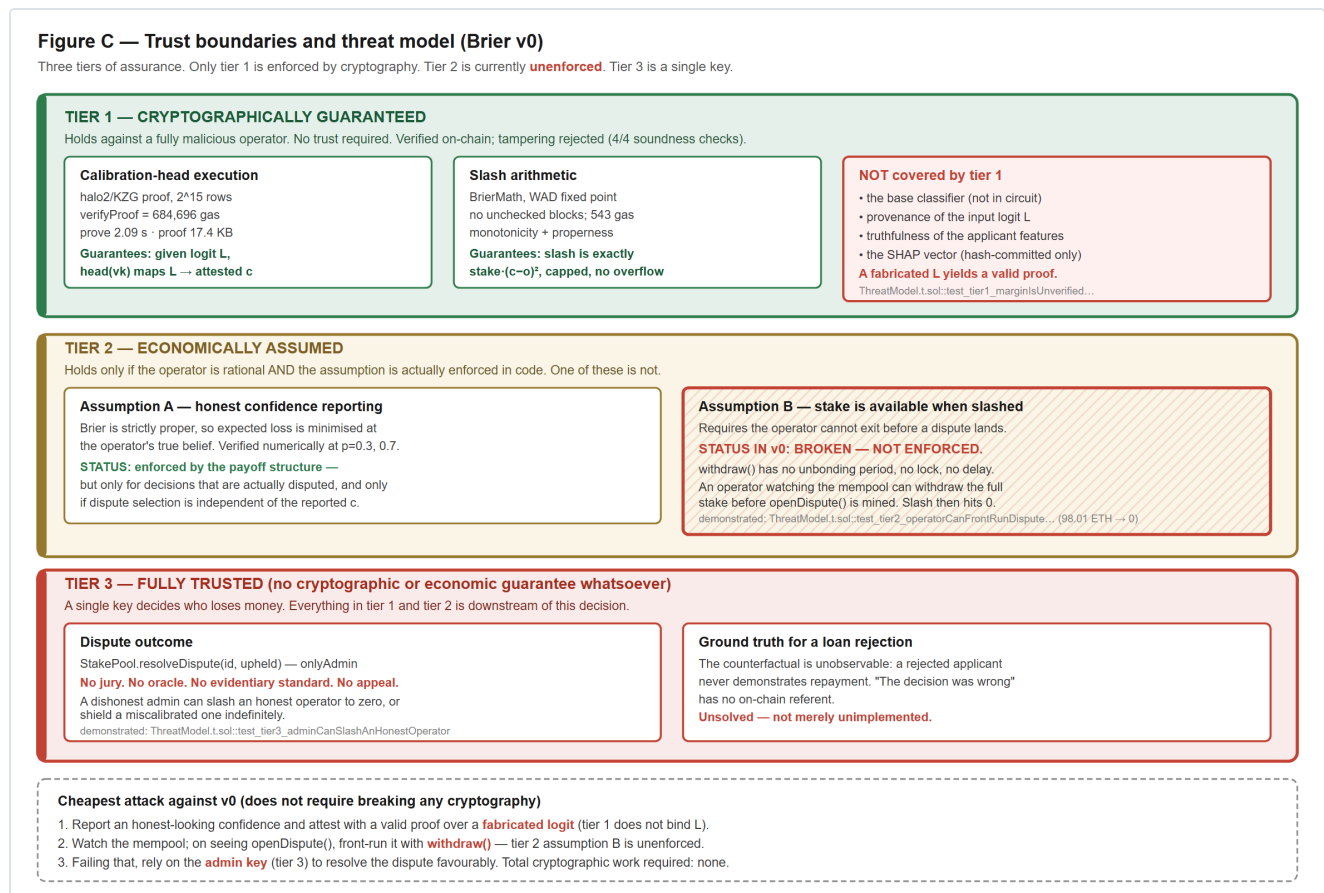


Figure C — Trust boundaries and threat model

This is the diagram against which every other claim in this document should be checked. Three tiers:

- **Tier 1 (cryptographically guaranteed, green).** Calibration-head execution and the slash arithmetic. Holds against a fully malicious operator.
- **Tier 2 (economically assumed, amber).** Assumption A (honest confidence reporting) is enforced by the payoff structure, conditionally. **Assumption B (stake is available when slashed) is BROKEN in v0** — hatched red in the figure — because `withdraw()` has no unbonding period.
- **Tier 3 (fully trusted, red).** Dispute outcome and ground truth. A single key decides who loses money.

Each weakness in the figure names the test that demonstrates it. The v0 attack path requires no cryptographic work: fabricate the logit (tier 1 does not bind it), or front-run the dispute with `withdraw()` (tier 2), or rely on the admin key (tier 3).

4.4 Figure D — Calibration reliability

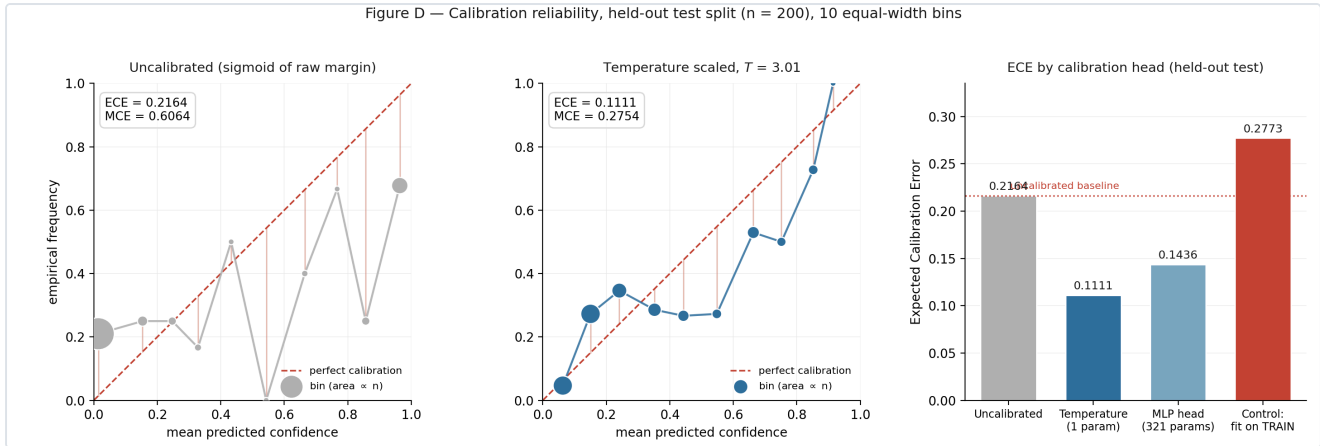


Figure D — Calibration reliability

Bin values are read directly from `artifacts/calibration/phase1_report.json`; no curve is fitted or smoothed. Left: uncalibrated, ECE 0.2164, with a bin at mean confidence 0.5430 whose empirical frequency is 0.0000. Centre: after temperature scaling with $T = 3.01$, ECE 0.1111. Right: ECE across heads, including the leakage control described in §5.2.

5. Methodology

5.1 Dataset and model

UCI Statlog German Credit: 1,000 real credit applications, 20 attributes, downloaded at build time. The label is inverted so that the positive class is *reject*. Attribute 9 (personal status and sex) is dropped, leaving 19 features (see §7.4). Split 60/20/20 into train (600) / calibration (200) / test (200), stratified, seed 42, with mutual disjointness asserted in code.

The base model is XGBoost (depth 8, 400 trees, learning rate 0.3, no regularisation) and is **deliberately overfit**: train accuracy 1.0000, test accuracy 0.7150. This is a methodological choice, not a tuned result. Demonstrating a calibration-insurance mechanism on an already-calibrated model would establish nothing; the overfit regime reproduces the overconfidence that makes calibration worth insuring. The base model's accuracy is not a claim of this work.

5.2 Calibration

Two heads were implemented and both are proved:

Head	Parameters	Form
Temperature scaling	1	$\sigma(L/T)$, T learned in log-space
MLP	321	$1 \rightarrow 16 \rightarrow 16 \rightarrow 1$, ReLU

Both emit a **logit**, with the sigmoid applied outside the module, so the circuit is affine + ReLU only and contains no transcendental function.

Both are fitted on the **calibration split only**. To show this matters rather than assert it, every run includes a control that fits the same head on the *training* split: that control yields $T = 0.1344$ and test ECE 0.2773, which is **worse than not calibrating at all** (0.2164). Fitting on training data *sharpens* an already-over-confident model.

5.3 Explainability

SHAP TreeExplainer over the base model, top-5 attributions per decision, in margin space. Correctness is checked by additivity (SHAP values plus base value must reconstruct the model margin; measured max absolute error 1.26×10^{-5}) and by bit-identical reproduction across reruns. Five directional claims, written before inspecting output, are verified by correlating each feature's value against its attribution toward reject.

5.4 Evaluation metrics

ECE (10 equal-width bins, weighted by bin population, empty bins skipped rather than counted as perfectly calibrated), MCE, Brier score, proof generation time, proof size, and on-chain verification gas. The ECE implementation is known-answer tested against hand-computed values.

5.5 Slashing validation

Monotonicity in miscalibration is verified deterministically at 101 confidence points and by fuzzing (256 runs per direction). Properness is verified numerically by scanning all 101 candidate reports at two distinct true probabilities, $p = 0.30$ and $p = 0.70$, and confirming the expected-loss minimiser equals p in both cases. Four scenarios are then run end-to-end on a local chain (§6.4).

6. Results

All figures below are produced by scripts in the repository. Environment: Windows 11, Python 3.13.5, xgboost 3.1.2, torch 2.9.1+cpu, EZKL 23.0.5, Foundry 1.7.1, laptop CPU, no GPU. Seed 42.

6.1R Calibration across 10 seeds (research pass)

Single-seed results are reported in §6.1 for reference; these are the figures the calibration claims rest on. Seeds are pinned in `config.EVAL_SEEDS`; the full list is always run.

Head	ECE mean $\pm$ std	Brier mean $\pm$ std	Accuracy mean $\pm$ std
Uncalibrated	0.1853 $\pm$ 0.0248	0.2060 $\pm$ 0.0191	0.7505 $\pm$ 0.0209
Temperature (1 param)	0.0870 $\pm$ 0.0218	0.1758 $\pm$ 0.0113	0.7505 $\pm$ 0.0209
MLP (321 params)	0.1055 $\pm$ 0.0234	0.1908 $\pm$ 0.0119	0.7330 $\pm$ 0.0199
<i>Control: fitted on TRAIN</i>	<i>0.2434 $\pm$ 0.0212</i>	—	—

Learned $T = 3.4657 \pm 0.4491$, and $T > 1$ in **every** seed. Calibration reduces ECE in **every** seed; the control is worse than not calibrating in **every** seed (paired Wilcoxon $p = 0.002$, the minimum attainable at $n = 10$).

Correction to a vo claim. vo asserted "temperature scaling beats the MLP head" from one run per head. Across seeds it holds as a *majority* result — 7/10 seeds, $W = 7.0$, $p = 0.03711$, median difference -0.01816 — not a uniform one. The claim is retained at that reduced strength and pinned by `tests/test_multiseed.py::test_temperature_beats_mlp_claim_matches_reported_strength`.

Cherry-picking audit. Seed 42, published alone in v0, ranks 6th of 10 by ECE reduction (48.7% vs 52.8% mean) while carrying the *highest* uncalibrated ECE of any seed. The single-seed figures were therefore pessimistic in absolute terms and middling in relative terms. `SEED = 42` was committed once, before results existed, and never changed; the repository contains no notebooks.

6.1 Calibration at seed 42 (single run, for reference)

Head	Params	ECE	MCE	Brier
Uncalibrated	—	0.2164	0.6064	0.2353
Temperature scaling	1	0.1111	0.2754	0.1897
MLP	321	0.1436	0.7766	0.2030
<i>Control: temperature fitted on TRAIN</i>	1	<i>0.2773</i>	—	—

ECE reduction: 48.7%. Learned $T = 3.0121$ ($T > 1$ confirms the base model required softening). Base accuracy: train 1.0000, test 0.7150.

Negative result, retained — and its strength corrected. At seed 42 the 321-parameter MLP head underperforms the 1-parameter head on held-out ECE (0.1436 vs 0.1111). Across the 10 pinned seeds (§6.1R) the effect is real but **not uniform**: temperature scaling has the lower ECE in **7 of 10** seeds (paired Wilcoxon $W = 7.0$, $p = 0.037$, median difference -0.018). With 200 calibration points the additional capacity usually fails to generalise, but in 3 of 10 seeds the MLP wins. The simplest head is both the better estimator on average and the cheaper circuit; that agreement is a measured outcome, not a design assumption, and it is a majority rather than a universal result.

6.2 Explainability

Additivity max absolute error 1.26×10^{-5} ; attributions bit-identical across 3 reruns; 5/5 directional sanity checks pass (`installment_rate_pct_income` $+0.621$, `duration_months` $+0.505$, `credit_history` -0.756 , `checking_status` -0.943 , `savings_status` -0.834).

6.3 Proving and verification

Metric	Temperature (1 param)	MLP (321 params)
Circuit size	$2^{15} = 32,768$ rows	$2^{15} = 32,768$ rows
Proving time	2.09 s	2.08 s
Native verification	0.094 s	0.058 s
Keygen	1.86 s	2.03 s
Proof size	17,447 B	17,566 B
Proving key	132 MiB	132 MiB
Calldata per proof	3,300 B	3,300 B
Tamper-soundness	4/4 rejected	4/4 rejected

Result, extended from 2 points to 10 (§6.3R). Proving cost is invariant to a **16,897× parameter increase**. A 10-point sweep from 1 to 16,897 parameters holds `logrows` = 15 at every point; rows used scale linearly with parameters (slope 0.98 rows/param, $r = 0.992$) while proving time does not follow (mean 2.13 s

± 0.09 , regression slope $+0.009$ s/decade, Spearman $ho = -0.188$, $p = 0.603$). Fixed lookup and column overhead dominate multiply-accumulate count. No fallback to a smaller head was required.

Scope limit. This holds *within* logrows = 15. The largest head fills 16,511 of 32,768 rows (50.4%), so the sweep does not cross the capacity boundary; where cost steps at logrows 16+ is **unmeasured**. The claim is "flat until the head stops fitting the circuit," not "flat without limit."

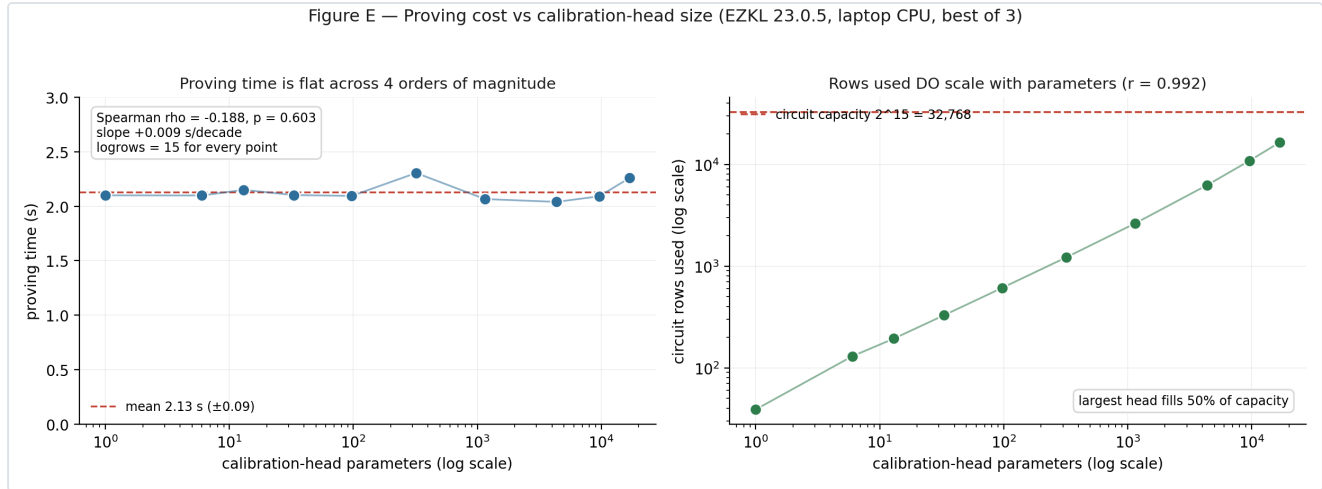


Figure E — proving cost vs head size

Soundness is tested rather than assumed: an honest proof verifies, while a tampered public output, a flipped byte at the proof head, a flipped byte mid-proof, and a wrong verifying key are each rejected.

6.4 On-chain cost and end-to-end behaviour

Operation	Gas
Halo2Verifier deployment (one-off)	2,942,192
verifyProof (real EZKL proof)	684,696
attest (verify + store)	887,376
openDispute	134,608
resolveDispute incl. slash + payout	97,390
stake	45,443
withdraw	29,616
BrierMath.slashAmount (pure)	543

Verification is $\sim 33\times$ a plain 21,000-gas transfer. The Brier arithmetic is 543 gas — approximately 0.08% of the per-decision cost. **Essentially all cost is halo2 verification, not the mechanism.**

End-to-end, on a local Anvil chain, with a real proof generated per decision:

Scenario	Confidence	Outcome	Slash (% of stake)	Prove
(a) confident + correct	0.9845	upheld	0.0239%	2.24 s
(b) confident + wrong	0.9367	overturned	87.7458%	2.25 s
(c) uncertain + wrong	0.5012	overturned	25.1173%	2.06 s

Ordering holds: (b) > (c) > (a). Confident-and-wrong costs $3,667\times$ confident-and-correct and $3.49\times$ uncertain-and-wrong. Percentages are of the stake remaining when each scenario ran — the three execute sequentially against one shrinking stake, so the percentage column is the comparable one and absolute ETH values across rows are not.

These figures assume the operator does not act on the vo exit path. Because `withdraw()` has no unbonding period, an operator that front-runs `openDispute()` reduces every slash in this table to zero; `ThreatModel.t.sol::test_tier2_operatorCanFrontRunDisputeByWithdrawing` executes exactly that, turning a 98.01 ETH slash into 0. The table reports what the mechanism computes, not what it can currently enforce (§7.3, Figure C tier 2).

6.5 Test suite

53 Solidity tests (Foundry) and 29 Python tests, all passing. This includes 5 tests in `ThreatModel.t.sol` that **demonstrate the weaknesses** in Figure C rather than assert the system is secure; a failure there means the threat model has drifted from the code.

7. Limitations and threat model

These are unresolved trust gaps in the present design. They are not future work.

7.1 Only the calibration head is proved

The zk proof establishes: *given a logit L committed on-chain, the head identified by this verifying key maps L to the attested confidence c .* It establishes nothing about where L came from. The base classifier is not in any circuit. **An operator that fabricates L obtains a proof that verifies**, as demonstrated by `test_tier1_marginIsUnverifiedOperatorSuppliedInput`, in which the chain accepts `type(int256).max` as a margin and still marks the attestation proved. Any description of this system as "zero-knowledge proving the AI decision" is false.

A secondary point: the proof attests the **quantised** computation. At input/param scale 13, the in-circuit fixed-point head differs from the float head by up to 4.2×10^{-4} . The on-chain confidence is the quantised one.

7.2 Dispute resolution is a single administrative key

`resolveDispute` is `onlyAdmin`. There is no jury, no oracle, no evidentiary standard, and no appeal. A dishonest administrator can slash a well-calibrated operator to zero (`test_tier3_adminCanSlashAnHonestOperator`) or shield a miscalibrated one (`test_tier3_adminCanShieldAMiscalibratedOperator`). Everything in tiers 1 and 2 is downstream of this key.

Beneath the implementation gap sits a harder problem: for a loan *rejection* the counterfactual is unobservable, because a rejected applicant never demonstrates repayment. "The decision was wrong" has no on-chain referent. This is unsolved, not merely unimplemented.

7.3 No unbonding period — tier 2 is broken

`withdraw()` checks only the caller's balance. There is no lock, delay, or pending-dispute check. An operator watching the mempool can withdraw the entire stake before `openDispute()` is mined. `test_tier2_operatorCanFrontRunDisputeByWithdrawing` executes this path: a decision that would otherwise incur a 98.01 ETH slash incurs 0, and the claimant receives nothing. The economic tier of Figure C is therefore not merely weak but unenforced, and the headline slashing results in §6.4 hold only against an operator that does not take this action.

7.4 Fairness

Attribute 9 (personal status and sex) is excluded from the feature set. This is a floor, not a fairness guarantee: proxy discrimination through correlated features is untested, no disparate-impact analysis was performed, and — most relevant to this mechanism — calibration is measured in aggregate and **not within protected subgroups**. A model can be well calibrated overall while badly miscalibrated for a subgroup, which is precisely the harm this mechanism should detect and currently does not.

7.5 Scope of the empirical claims

Results are from a single dataset ($n = 1,000$; test split $n = 200$), one seed, one model family, and one decision class, on a local devnet. Calibration drift over time is not evaluated. Staking parameters are demonstration values with no actuarial basis, and correlated tail risk — one bad model version producing thousands of simultaneous claims — is not modelled.

8. Future work

1. **Decentralised adjudication.** Replace the admin key with a Kleros-style juror pool [7], adding an evidentiary standard, appeals, and economic security exceeding the value at risk. Requires resolving what evidence establishes that a rejection was wrong given the unobservable counterfactual (§7.2).
 2. **Unbonding period.** Close §7.3 with a withdrawal delay exceeding the dispute window, and analyse the resulting capital-efficiency cost to the operator.
 3. **Input-logit provenance.** Three paths, in decreasing order of guarantee strength: prove the base classifier in-circuit (expensive for a depth-8, 400-tree ensemble); commit to feature vector and model weights so a fabricated logit is *detectable* though not *proved*; or attest the base model through a secure enclave, substituting a hardware trust assumption for a cryptographic one. Only the third is cheap today, and it is strictly the weakest.
 4. **Collusion detection.** Graph-based analysis over the claimant/operator bipartite graph to detect self-dealing disputes. Untried here; listed as a direction, not a result.
 5. **Subgroup calibration as the slashed quantity.** Slashing on within-group ECE rather than aggregate confidence would target the harm in §7.4 directly.
 6. **Non-deterministic decision classes (LLM outputs).** We flag this as **substantially harder, not a natural extension**. The present design depends on a deterministic decision with a scalar confidence and a binary outcome. Free-form generation has no canonical scalar confidence, no binary ground truth, and no stable notion of "the same decision" across sampling seeds. Applying a proper scoring rule requires first defining the event space being scored, which is an open problem rather than an engineering task.
-

9. Conclusion

We have demonstrated a working, measured prototype of confidence-weighted slashing: a strictly proper scoring rule implemented in fixed-point Solidity with monotonicity and properness verified numerically, driven by a calibration step that is proved in zero knowledge in 2.09 s and verified on-chain for 684,696 gas, on real credit data where calibration reduces ECE from 0.2164 to 0.1111. We have **not** demonstrated a trust-minimised protocol: only the calibration head is proved and its input logit is unverified, dispute resolution is a single administrative key, and the absence of an unbonding period lets an operator front-run a dispute and reduce a 98% slash to zero. The mechanism-design claim is supported by the evidence presented; the systems claim is not, and closing that distance is the substance of the work that remains.

References

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- [2] Centre for Business and Commercial Laws, NLIU, *Analysing RBI's Digital Lending Directions 2025: A Positive Step Towards Responsible Lending?* — documenting that the Working Group's algorithmic-audit and transparency recommendations were incorporated into neither the 2022 Guidelines nor the 2025 Directions. <https://cbcl.nliu.ac.in/contemporary-issues/analysing-rbis-digital-lending-directions-2025-a-positive-step-towards-responsible-lending/>
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Appendix: reproducing the results

```

pip install -r requirements.txt
bash scripts/00_setup_tools.sh          # EZKL CLI v23.0.5

python scripts/10_train_calibrate.py    # §6.1 (ECE gate; fails loudly if not reduced)
python scripts/20_explain.py            # §6.2
python scripts/30_export_onnx.py        # ONNX export, fidelity-checked
python scripts/31_zk_prove.py           # §6.3 circuits, proofs, soundness
cd contracts && forge test               # §6.5 53 tests incl. ThreatModel.t.sol
python -m pytest tests/ -q              # §6.5 29 tests

anvil &                                 # §6.4 end-to-end
forge script script/Deploy.s.sol:Deploy --rpc-url http://127.0.0.1:8545 --broadcast
python scripts/40_demo_e2e.py
python scripts/90_render_results.py      # regenerate RESULTS.md
python scripts/91_figure_d_reliability.py # Figure D
python scripts/92_render_svg.py          # Figures A and C

```